

Rolando Miranda, BS.

NYU Tandon Mechanical Engineering Graduate Student

Phone: +1 (646) 922 2020

Email: rolomiranda98@gmail.com

Portfolio: <https://rolando-miranda.github.io/>

Based in Bronx, NY, open to relocation after graduation.

Background and Summary

Mechanical Engineer with experience in automated process development, high-volume manufacturing and coursework in mechatronics. Combines technical expertise in mechatronics, mechanical engineering, project management and lean manufacturing. Professional background in Class III medical device manufacturing, compliance with quality standards, test method design and validation. At Terumo-Neuro, managed validation efforts across multiple product lines, contributing to \$3M backorder reduction and doubling production capacity for the highest-volume microcatheter. Currently functioning as a researcher supporting design and automatic control of electrochemical reactors.

Education

- **NYU Tandon School of Engineering:** MS. in Mechanical Engineering. Coursework: : Mechatronics, Linear Control, Applied Math, Thermal Engineering, Autonomous Vehicles & Solid Mechanics, Reinforcement Learning, Robot Perception.
Graduation: May 2027
- **Latin American University of Science and Technology (ULACIT):** Master's in Project Management. Graduated Suma Cum Laude (September 2025) Completed while working full-time at Terumo-Neuro
- **Universidad de Costa Rica:** Licenciature's with Honors in Mechanical Engineering. GPA 9.71 out of 10. Graduated in April 2023
- **Universidad de Costa Rica:** Bachelor's with Honors in Mechanical Engineering. GPA 9.71 out of 10. Graduated in April 2021

Work Experience

NYU Tandon School of Engineering, **Research Assistant, Bui Lab (Electrochemistry)** (5/2026 - current)

- Design and build laboratory automation platforms for CO2 and NO3 electrolysis
- Deliver power supply, mechanical and electronic subsystem design for automated reactor actuation.
- Transform electrochemical process requirements into technical design specifications and features.
- Build software for actuation control, sensor reading and process orchestration

NYU Tandon School of Engineering, **Graduate Course Assistant, Thermo-fluids Laboratory** (9/2025 – 5/2026)

- Designed and built equipment for thermal system evaluation for channel flow applications from scratch for a new laboratory.
- Assist students in their projects related to thermal system modeling, improvement and testing.
- Draft laboratory SOPs, run laboratories and ensure correct practice execution.

Terumo-Neuro, **Manufacturing Engineer II** (6/2023 – 7/2025):

- Managed cross-functional validation project that doubled production capacity for highest-volume microcatheter, coordinating R&D, QA, Manufacturing, and Regulatory teams through qualification protocol development and execution.
- Executed raw material validation strategy across five product lines partnering with R&D, reducing \$3M in backorders through systematic design verification (DV), process validation (PV) and stakeholder alignment.
- Exceeded corporate yield targets across assigned product lines by standardizing SOPs and aligning production, quality, and management teams on process improvements.

MicroVention-Terumo, **Manufacturing Engineer I** (9/2021 – 6/2023)

- Performed hands-on functional testing of Class III medical devices for design & process validation.
- Authored validation documentation collaborating across R&D, QA, and Manufacturing to achieve process qualification.

MicroVention-Terumo, **Manufacturing Engineering Intern** (5/2021 – 9/2021)

- Led small transfer projects due to line relocation.
- Collaborated in Root Cause Analysis (RCA) for manufacturing issues.
- Responsible of high-value raw material scrap and handling.

Junior Engineer, **CIRE-TOP Group Engineering Consulting Firm** (1/2021 – 5-2021)

- Ensured safety of high-pressure steam boilers with in-field steel thickness measurements and pressure vessel solid mechanics. Executed design tasks for piping systems for energy and mining applications.

Key Skills

- **Technical Skills:**
 - Software:** Mathematical simulation and analysis: MATLAB, Simulink, Minitab for statistics.
CAD: High competency in SolidWorks, AutoCAD, Fusion360.
Microsoft Suite: Competency on all MS Office tools, MS Project, Power BI.
 - Manufacturing:** GxP compliance, process validation, FMEA, Class III medical device manufacturing, PTC Windchill.
 - Working knowledge of reductive and additive manufacturing techniques, molding and applied material science.
 - Engineering:** Statistics for process validation, FMEA, process automation, FEA, mechanical design.
 - Prototyping:** Arduino, embedded systems, 3D printing, electronic circuit design and mechatronic integration.
 - Programming:** Computation, visualization and automation with Python. Familiar with C for embedded systems.
 - AI Tools:** Claude Code and coding agents to accelerate scripting and debugging, LLM tools for technical writing, research synthesis, and process documentation.
- **Methodologies:** **Project Management:** Agile, Waterfall, Risk Management
Quality: Lean Manufacturing, Six Sigma, Root Cause Analysis
- **Languages:** English Fluency (TOEFL 105), Native Spanish, German (Basic A2)